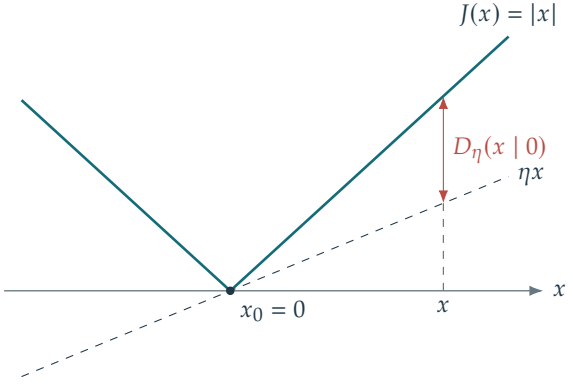

$$|\eta| < 1$$

$$\begin{aligned} D_\eta(x \mid 0) &= |x| - \eta x \\ &\geq (1 - |\eta|)|x| \end{aligned}$$

The margin $1 - |\eta|$ is positive precisely when the subgradient does not saturate.